SANABHI GAURAW

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EDUCATION

B.S. Electrical Engineering & Computer Science, B.A. Physics August 2026 – May 2029 (Expected)
University of California, Berkeley, Berkeley, CA

Plano East Senior High School, Plano, TX Graduated May 2026

- Rank: 4/1300 SAT: 1560 (770/790)

RESEARCH

Acoustic Wave Propagation Research (NSF EFRI NewLaw Grant) – University of North Texas

Principal Investigator: Dr. Arkadii Krokhin

· Developed a COMSOL model of acoustic pulse propagation through glycerine and other high-viscosity fluids.
· Designed a fluid-embedded test system for measuring acoustic properties for non-destructive imaging, inspired by prior work in elastic bulk modulus elastography.

Scintillator Calibration – Lawrence Berkeley National Laboratory (URAP) August 2026 - Present

Principal Investigator: Dr. Bethany Goldblum

· Calibrated a scintillator detector for the LBNL nuclear physics lab, using radioactive sources to measure energy deposition and light yield.
· Developed ROOT macros to automate data collection and analysis, improving efficiency and accuracy of fit of simulated data onto experimental data.

PROJECTS

PHYSICA-X: Web Toolkit for Interactive Physics Lessons | Bun, PostgreSQL, TypeScript

· Built and deployed a frontend and backend that lets registered teachers create and publish simulation-based lessons to a shared database accessible to any student.

RTL-to-Layout ASIC Implementation | Verilog, Open-Source EDA

· Designed and synthesized Random Number Generator in Verilog, taking it from RTL through technology mapping to physical implementation with an open-source EDA toolchain.
· Analyzed timing reports to find critical paths and evaluate setup-time constraints and timing-vs-area trade-offs.
· Compared two implementation configurations, documented changes in critical-path delay and power consumption.

CPU Cache and Performance Simulator | C++

· Built a cache simulator with configurable capacity, associativity, line size, and replacement policies (LRU, FIFO).
· Writing a trace-driven harness to measure hit/miss rates across workloads.

TECHNICAL SKILLS

Languages	C++, Python, Verilog, CUDA, VHDL
Tools	ROOT, COMSOL, open-source EDA flow, LaTeX, MATLAB, OpenMP

HONORS & ACTIVITIES

- National Science Bowl, High School Team Captain Mar. 2024, 2025, 2026
- USNCO, USAPhO, AIME Qualifier Mar. 2025
- FIRST Robotics Competition Global Qualifier, FTC UIL State Champions Mar. 2025, Apr. 2024